

OpsIntel

AI Infrastructure Incident Investigator powered by Exasol

Telemetry	Exasol Personal	AI Investigator	OpsIntel Dashboard
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The Problem

- Infrastructure incidents produce scattered metrics, alerts, and timeline events.
- Operators must correlate CPU, memory, latency, errors, HTTP failures, and events under time pressure.
- AI answers can be fast but risky when they invent metrics or confuse correlation with causation.
- OpsIntel makes Exasol the evidence layer and constrains AI to the retrieved evidence.

How OpsIntel Works

- 1. Synthetic infrastructure telemetry and incident events are stored in Exasol Personal.
- 2. Exasol SQL calculates baselines, incident values, anomaly flags, and severity.
- 3. A read-only SQL safety gate validates queries before execution.
- 4. The AI investigator receives structured evidence plus correlated timeline events.
- 5. Streamlit presents the investigation, evidence, timings, and incident timeline.

Exasol Is the Evidence Engine

- OPSINTEL.SERVICES — service inventory and ownership context.
- OPSINTEL.SERVICE_METRICS — 50,000 synthetic telemetry rows.
- OPSINTEL.INCIDENT_EVENTS — 5 correlated incident/recovery events.
- Measured investigation output: anomaly score 5, severity CRITICAL, confidence HIGH.
- Exasol is not just storage: SQL performs the analytical correlation that drives the investigation.

Example: web-03 Incident

- CPU: 41.96% baseline → 93.86% incident.
- Memory: 57.91% → 81.75%.
- Response time: 180.02 ms → 1,252.12 ms (6.96×).
- Error rate: 0.41% → 17.42% (42.47×).
- HTTP 5xx: 2.62 → 184.44 (70.32×).
- Correlated timeline: warning → connection-pool saturation → 5xx spike → latency breach → recovery.

AI + Safety + Performance

- Evidence-first prompt rules prevent invented metrics and unsupported certainty.
- Observation and inference are explicitly separated; the report labels the connection-pool explanation as a hypothesis.
- SQL safety blocks destructive operations such as DROP, DELETE, UPDATE, INSERT, ALTER, and TRUNCATE.
- The dashboard measures Exasol time, AI time, and total investigation time.
- A real dashboard run recorded Exasol analytics at 514.4 ms and total investigation time at 25,066.9 ms.

Impact & Next Steps

- Reduce manual incident correlation from a multi-source investigation to one evidence-backed workflow.
- Give operators a reproducible timeline, quantified anomalies, and a cautious AI explanation.
- Current MVP: Exasol analytics + anomaly detection + AI investigation + safety gate + Streamlit dashboard.
- Next: live telemetry ingestion, broader service coverage, alert integration, approval-based remediation, and richer forecasting.
- Repository: github.com/rahulkhairnar/opsintel